



RETURNSHIELD AI

Return fraud decisioning for modern commerce

PURPOSE-BUILT FOR POST-PURCHASE RISK

Stop return fraud before it becomes refund loss.

Score every return request, explain the decision, and block risky refunds before cash leaves the system.

WHAT IT DELIVERS

0-100

Risk score

3

Decision
bands

Explain

Reason trace

Live

API scoring

Adopt fast

Score, explain, review, and learn in one control loop.

Merchant value

- Fewer wrongful refunds
- Faster review for good returns
- Better analyst consistency

System output

- Risk score from `0` to `100`
- `AUTO\_APPROVE`, `MANUAL\_REVIEW`, or `HOLD\_REFUND`
- Reason codes and readable explanations

Fast decisions with visible reasoning. Built for merchants who need control, not opaque scores.

OPERATIONAL SNAPSHOT

Score

0-100 risk scale

Explain

reason codes

Learn

feedback loop

HOLD

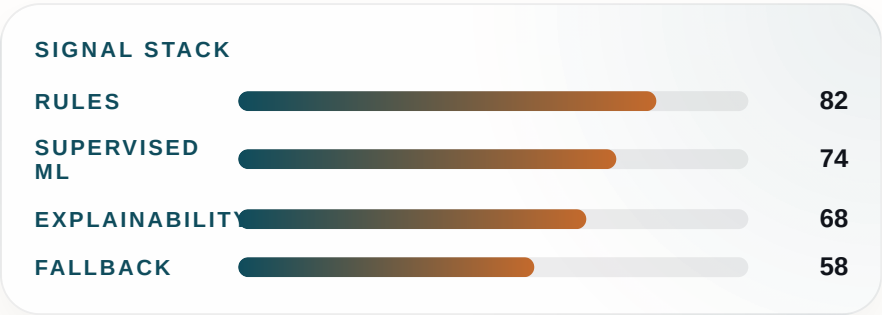
HIGH-RISK CASES

Built for the fraud surface checkout tools miss: empty-box claims, wardrobing, item swaps, fraud rings, and suspicious return narratives.

RETURNSHIELD AI

Product features

ReturnShield AI works as an operational fraud console, not just a scoring endpoint.



Return scoring

Scores every return before the refund is released.

Explainability

Shows why a request was flagged in plain language.

Case review

Routes medium and high-risk returns to analysts.

Feedback loop

Stores analyst decisions for retraining and follow-up.

Fraud ring visibility

Surfaces connected accounts and repeat behavior.

Reason codes

Turns score output into actions the team can trust.

TYPICAL WORKFLOW

1. Return request arrives.
2. Payload is normalized.
3. Rules and models score the case.
4. The return is approved, reviewed, or held.

Algorithm and accuracy

The live path is rule-led supervised ML. NLP, graph, and anomaly signals stay available for explanation and manual review.

LAYER	SIGNAL	ROLE
Rules	Policy checks	Fast guardrails for known patterns
Supervised ML	Customer, order, shipment, return fields	Finds hidden tabular patterns
NLP / text	Return reasons and support text	Catches suspicious language shifts
Graph / anomaly	Outliers and unusual combinations	Flags abnormal behavior

$$\text{final_score} = (\text{rule_score} * 0.35) + (\text{ml_score} * 0.65)$$

Fallback mode keeps the API responsive when the promoted model artifact is unavailable.

0-39

AUTO_APPROVE

Low risk, release the refund.

5. Analyst feedback is stored for improvement.

DECISION LOOP

REQUEST

SCORE

REVIEW

LEARN

40-69

MANUAL_REVIEW Needs analyst inspection.

70-100

HOLD_REFUND

Strong fraud indicators present.

ACCURACY STORY

- PR-AUC for model selection
- Recall and precision on fraud cases
- F1 and false positive rate for tie-breaks
- Reviewability for analysts and auditors

The repo tracks precision, recall, F1, ROC-AUC, and PR-AUC in training runs, but it does not publish audited benchmark percentages. That keeps the pitch credible.